Extraocular Muscles

Images used: 11.10, 10.18, 11.17, 11.18, 11.15, 11.16, 11.19, 11.20, 13.14, 13.17

I. Macroscopic anatomy of the extraocular muscles

A. Six extraocular muscles:

1. Medial rectus (MR)
2. Lateral rectus (LR)
3. Superior rectus (SR)
4. Inferior rectus (IR)
5. Superior oblique (SO)
6. Inferior oblique (IO)

B. Origin of the rectus muscles

1. Common tendinous ring aka annulus of Zinn

2. The origin of the MR and SR also attach to the dural sheath of the optic nerve

a. Clinical correlate: Retrobulbar optic neuritis

C. Insertions of the rectus muscles

1. The four rectus muscles insert into the anterior globe

2. Tendons pierce Tenon capsule and merge with the scleral fibers.

D. Extraocular muscle anatomy

1. Medial rectus

    a. Origin:

    b. Insertion:

    c. Innervation:

2. Lateral rectus

    a. Origin:

    b. Insertion:

    c. Innervation:

3. Superior rectus

    a. Origin:

    b. Insertion:

    c. Innervation:

4. Inferior rectus

    a. Origin:

    b. Insertion:

    c. Innervation:

5. Superior oblique muscle

    a. Pathway

    b. Origin:

    c. Insertion:

    d. Innervation:

6. Inferior oblique

    a. Pathway

    b. Origin:

    c. Insertion:

    d. Innervation:

II. Extraocular muscle movement definitions

A. Fick's axes DRAW

1. X axis

    a. Horizontal
    b. Runs nasal to temporal

2. Y axis

    a. Horizontal
    b. Runs anterior to posterior

3. Z axis
    a. Vertical
    b. Superior to inferior

B. Ductions

1. Rotation around the vertical z axis

    a. Medially:

    b. Laterally:

2. Rotation around the horizontal x axis

    a. Elevation:

    b. Depression:

3. Rotation around the horizontal y axis

    a. Rotation nasally:

    b. Rotation temporally:

C. Position of gaze

1. Primary gaze:

2. Secondary gaze:

3. Tertiary position:

III. Extraocular muscle movements

A. Movement from primary position

1. Horizontal rectus muscles only have one action

a. Lateral rectus contraction

b. Medial rectus contraction

2. Vertical rectus muscles

a. Superior rectus

(1) Primary action:

(2) Secondary actions:

b. Inferior rectus

(1) Primary action:

(2) Secondary actions:

3. Oblique muscles

    a. Superior oblique

        (1) Primary action:

        (2) Secondary action:

    b. Inferior oblique

        (1) Primary action:

        (2) Secondary action:

B. Movement from secondary positions

1. Vertical rectus muscles

a. Abduction

(1) There will be no torsional movement, and the eye is already abducted due to the LR.

(2) SR will cause

(3) IR will cause

b. Adduction

(1) SR will cause

(2) IR will cause

2. Oblique muscles

    a. Abduction

        (1) SO will cause

        (2) IO will cause

    b. Adduction

        (1) SO will cause

        (2) IO will cause

3. Clinical correlate: Extraocular muscle assessment

    a. Horizontal eye movements

        (1) If the eye does not abduct:

        (2) If the eye does not adduct:

    b. Vertical eye movements

        (1) Adducted position

            i. If the eye does not elevate:

            ii. If the eye does not depress:

        (2) Abducted position

            i. If the eye does not elevate:

            ii. If the eye does not depress:

IV. Muscle coordination

A. Synergist/agonist muscles work together to accomplish the same action.

B. Antagonist muscles work in opposite directions.

1. Sherrington's law of reciprocal innervation: contraction of one muscle is accompanied by a simultaneous and proportional relaxation of the antagonist muscle.

C. Yoke muscles: the muscles of the two eyes that work together to cause symmetric, binocular eye movements.

1. Hering's law of equal innervation: innervation to the muscles of the two eyes is equal and simultaneous.